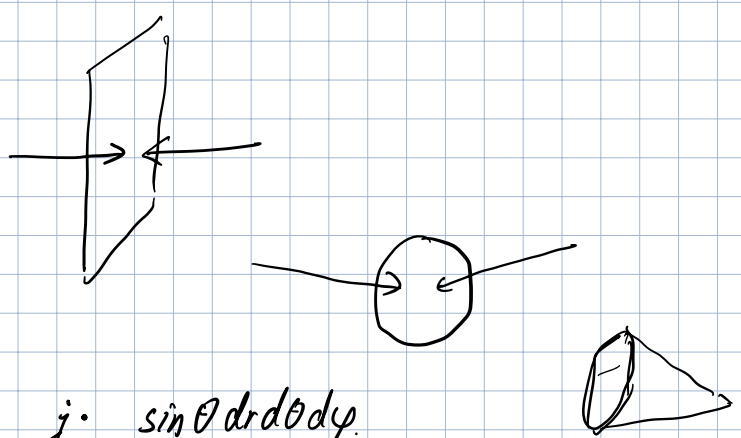


$$\frac{nS_1}{S_2} = \frac{n_\cdot}{\cos\theta}$$

(1) $\frac{1}{\cos 20^\circ}$ $\sec 20^\circ$

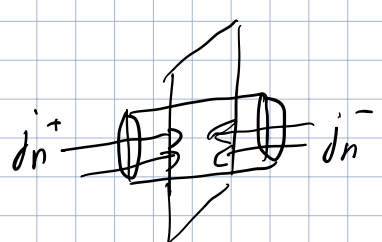
(2) $1:1$



$j \cdot \sin\theta \, dr d\theta d\varphi$

$j \cdot \underset{\sim}{dS}$ $j \cdot$

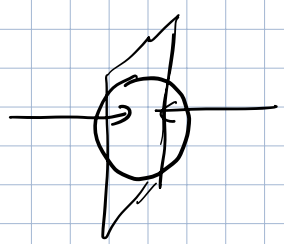
$$\frac{d}{dt}\left(\frac{dNe}{da}\right) = \frac{d}{da}\left(\frac{dNe}{dt}\right) = \frac{dj}{da}.$$

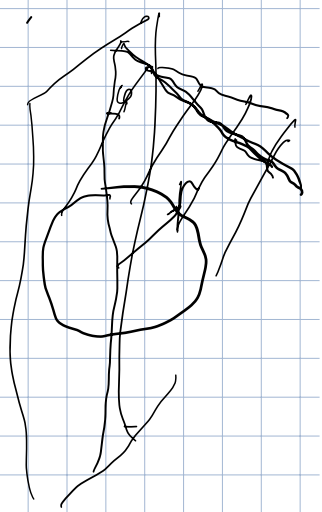


$dn^+ + dn^-$ $\quad \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$

$\frac{1}{\varepsilon_0} \int \rho \, dV$





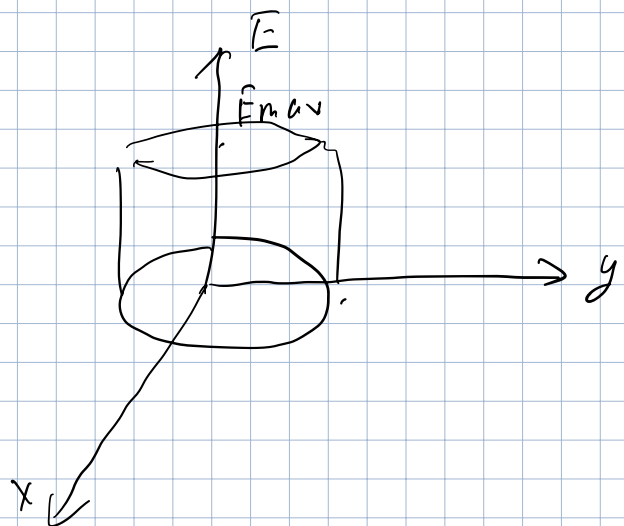


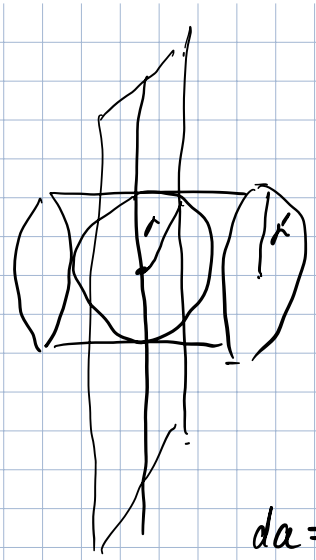
$dn^+ + dn^- =$

$\varphi(r, \theta, \varphi)$.

$$\int_0^{\pi} \varphi' \sin\phi \cos\theta \, d\phi = dn^+ + dn^-$$

$$\varphi = \int_0^{\pi} \int_0^{2\pi} \varphi' \sin\phi \cos\theta \, d\phi \, d\theta.$$





$$\frac{\frac{4V}{S}dN}{dV} =$$

$4\pi r^2$

$da = d\frac{1}{3}\pi r^2$

$= \frac{1}{3}\pi r dr$

$8\pi r\,dr$

$$\frac{4V}{S}\frac{dN}{d\frac{4}{3}\pi r^3} = \frac{V}{S}\frac{2dN}{2\pi r^2 dr} = \frac{V}{S}\frac{2dN}{r\,da}$$

$\frac{4}{3}\pi r^3$

$\frac{4}{3}\pi r^3$

$\frac{2}{3}\frac{dN}{da}$

$\frac{\frac{4}{3}\pi r^3}{4\pi r^2}$

$\frac{8}{3}\frac{dN}{da}$

$\frac{1}{3}r$

$\frac{6V}{S}$

$$\bar{L} = \frac{\int_{-r}^{r} x\sqrt{r^2-x^2}\,dx}{r} =$$

πr^2

$4\pi r^2$